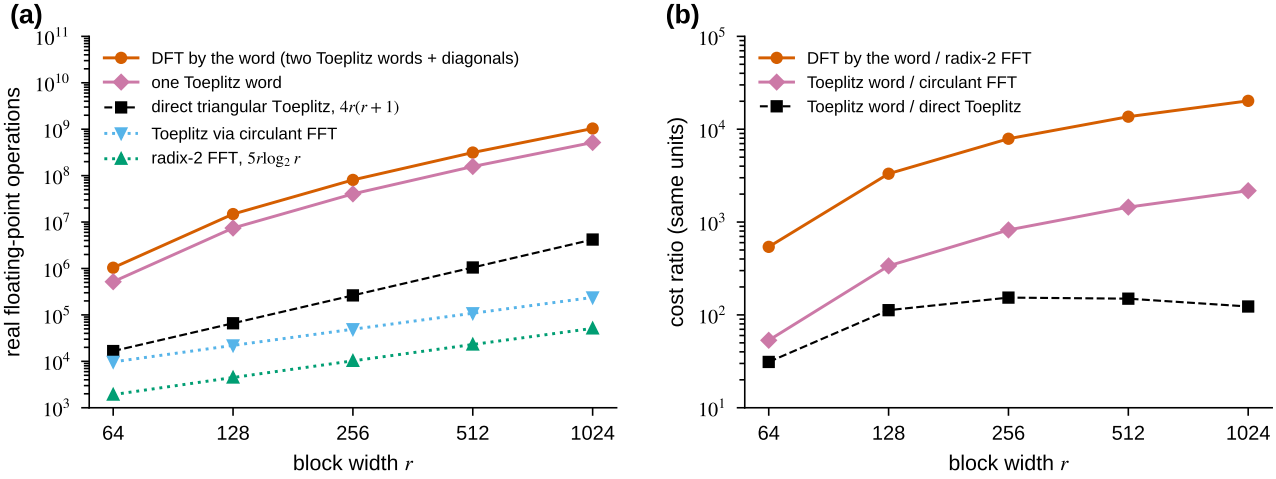
(a)
real floating-point operations
$10^{11}$
$10^{10}$
$10^{9}$
$10^{8}$
$10^{7}$
$10^{6}$
$10^{5}$
$10^{4}$
$10^{3}$
DFT by the word (two Toeplitz words + diagonals)
one Toeplitz word
direct triangular Toeplitz, $4r(r+1)$
Toeplitz via circulant FFT
radix-2 FFT, $5r\log_2 r$
64
128
256
512
1024
block width $r$
(b)
cost ratio (same units)
$10^{5}$
$10^{4}$
$10^{3}$
$10^{2}$
$10^{1}$
DFT by the word / radix-2 FFT
Toeplitz word / circulant FFT
Toeplitz word / direct Toeplitz
64
128
256
512
1024
block width $r$